

A coupled disease-misinformation model of measles transmission in the Canadian context

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ABSTRACT

Infectious disease dynamics are increasingly shaped not only by biological processes but also by the spread of misinformation. This study presents a coupled disease-misinformation model, SMIRK, to evaluate the impact of misinformation on a recent measles outbreak in Canada. The novel model combines a standard SIR framework for measles transmission with an SIS-like KMK model for misinformation spread. Two misinformation recovery paradigms are explored: constant-rate recovery and recovery induced by contact with infected individuals. Parameters were estimated using least-squares fitting to epidemiological data from Health Canada and Public Health Ontario (December 2024 to March 2025). Model calibration suggests that misinformation rapidly reaches a stable equilibrium, effectively saturating the population regardless of recovery paradigm. Due to the rapid information homogenization of the population, measles is predicted to behave as a standard low- R_0 infection under most calibrations. Despite its limitations, the SMIRK model offers a proof of concept for integrating misinformation into epidemiological models, underscoring the need for more nuanced modeling of informational dynamics in public health forecasting.

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1. Introduction

In an era of rapid information exchange, the spread of infectious diseases is no longer determined solely by biological factors but also by the circulation of disinformation and misinformation. False or misleading information undermine public health efforts and impact infectious disease dynamics. For instance, misinformation impacts vaccination uptake (Skafle et al., 2022), prophylaxis compliance (e.g., mask or condom use, isolation) (Bogart et al., 2011; Stein et al., 2021; Wang et al., 2021), and public policy (Glenza, 2025), and has been credibly linked to the resurgence of certain diseases and worsening of epidemics (Caceres et al., 2022; Novilla et al., 2023).

Specifically, dis- and misinformation have led to vaccine hesitancy and thus the resurgence of measles in North America (Novilla et al., 2023). Measles is a highly contagious, vaccine-preventable airborne disease. In Ontario, it mostly infects infants, children, and adolescents, who are also at highest risk of complication and hospitalization, though the majority of cases are not severe and non-fatal (Public Health Ontario, 2025). Though measles immunization is required to attend

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primary and secondary school in Ontario, there are exceptions for medical reasons or due to conscience or religious belief (Government of Public Health Ontario, 2025). The other province driving the national outbreak is Alberta, which does not mandate vaccinations for school enrollment (Alberta Health Services, 2025). Though anti-MMR (measles, mumps, and rubella) vaccine claims are not new, the COVID-19 pandemic has exacerbated anti-vaccine sentiment generally, with downstream effects on compliance (Skafle et al., 2022). Because of its very high basic reproductive number ($12 < R_0 < 18$), i.e., the expected number of cases arising from a single infectious individual in a fully susceptible population, measles is particularly sensitive to small drops in vaccine coverage and thus to vaccine hesitancy, which has led to outbreaks in Canada and the United States, despite both countries declaring the disease eliminated in 1998 and 2000, respectively (Novilla et al., 2023). This includes a recent outbreak in Canada concentrated in southern Ontario and driven by unvaccinated individuals (Zafar, 2025).

Despite the macro-level understanding of the links between disease and misinformation, there is a dearth of research on micro-level impact on outbreaks and epidemic curves. This paper models the interaction between misinformation and infectious disease, specifically measles. Through a novel multiplicative composite model using an SIR model for measles and SIS model for misinformation, this paper models the synergy of misinformation and disease spread and aims to fill the gap in understanding of the natural history of misinformation and how its dynamics affect epidemics, specifically characterizing the current Canadian measles outbreak.

2. Methods

2.1. Modelling approach

The proliferation of false information can be modelled as that of an infection. Though network approaches are often used to take advantage of the inherent network structure of social media (Butts et al., 2023; Govindankutty & Gopalan, 2024; Prandi & Primiero, 2020; Rabb et al., 2022; Sulis & Tambuscio, 2020), compartmental models are also popular (Gavric & Bagdasaryan, 2019; Panchal et al., 2022). Here, an SIS-analogous approach is proposed (Fig. 1). Under this KMK model, individuals are either knowledgeable (K) or misinformed (M). The knowledgeable are “infected” with misinformation proportionally to the transmission rate ϕ and the misinformed proportion of the population; they “recover” at rate λ . Simple, constant “recovery” (A in Fig. 1) could take place via loss of interest in the misinformation or via cumulative contact with accurate public health information from trusted sources in the absence of contact with misinformed individuals. Alternately, the misinformed could be driven back to the knowledgeable state via contact with those infected with the disease about which they are misinformed, due to fear of infection or direct contact with trusted medical professionals (B in Fig. 1). This model makes a number of simplifying assumptions, including that there is neither a misinformation-immune state (representing, e.g., healthcare professionals or their close contacts) nor an absorbing disinforming state and that the population is well-mixed. Moreover, the model assumes uniform information contacts, which obscures the very lopsided media and (mis)information landscape and the presence of misinformation “superspreaders” (Baribi-Bartov et al., 2024; DeVerna et al., 2024).

A parallel model of disease dynamics is adopted (Fig. 2). Specifically, a dynamic SIR (susceptible - infected - removed) approach with birth and death rate μ , transmission rate β , recovery rate γ , and vaccination rate ρ is proposed to balance the natural history of measles, which has previously been modelled using the SIR approach (Arsal et al., 2020; Bjørnstad et al., 2002; Eilertson et al., 2019), and simplicity for the sake of modelling. This model assumes a well-mixed population.

These models are combined in Fig. 3 with parameters listed in Table 1. In the multiplicative composite SMIRK paradigm, knowledgeable and misinformed infectious individuals have distinct transmission rates (β_k and β_m , respectively) and misinformed individuals are assumed to not vaccinate. Two cases of becoming knowledgeable (“recovering” from misinformation) are considered: recovery at a constant rate λ_1 and recovery due to contact with a disease-infected person with “transmission” rate λ_2 . These are marked as A and B, respectively, in Figs. 1 and 3. Governing systems of differential equations for these systems are shown in Equations (1) and (2) in the Appendix. Notable assumptions of this model are that misinformed individuals do not vaccinate, the force of misinformation infection and misinformation recovery rate are the same for all disease states, that information state is the salient division in the population (as opposed to, e.g., age), and individuals enter the population either knowledgeable or misinformed proportionally to the current population.

From estimates of disease transmission rates and the well-established infection duration for measles (γ), we find the effective reproductive number for the knowledgeable (R_k) and misinformed (R_m), both unitless. The effective reproduction number is the number of individuals infected by a single case (either knowledgeable or misinformed) in a partially

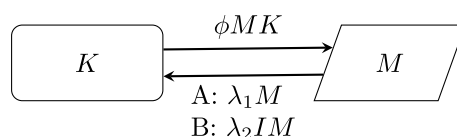


Fig. 1. Stock and flow diagram of KMK misinformation model. Misinformation “recovery” occurs either by simple constant recovery rate (A) or through contact with individuals infected with the disease about which the population is misinformed (B).

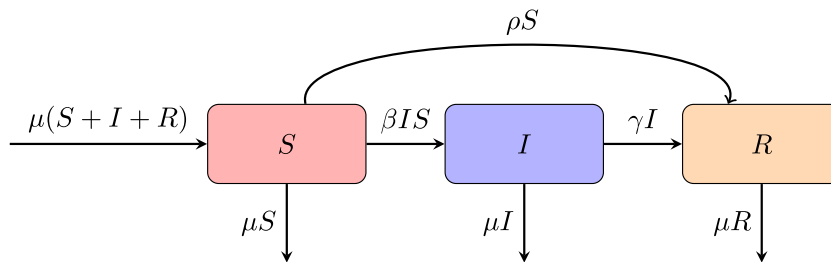


Fig. 2. Stock and flow diagram of SIR measles model.

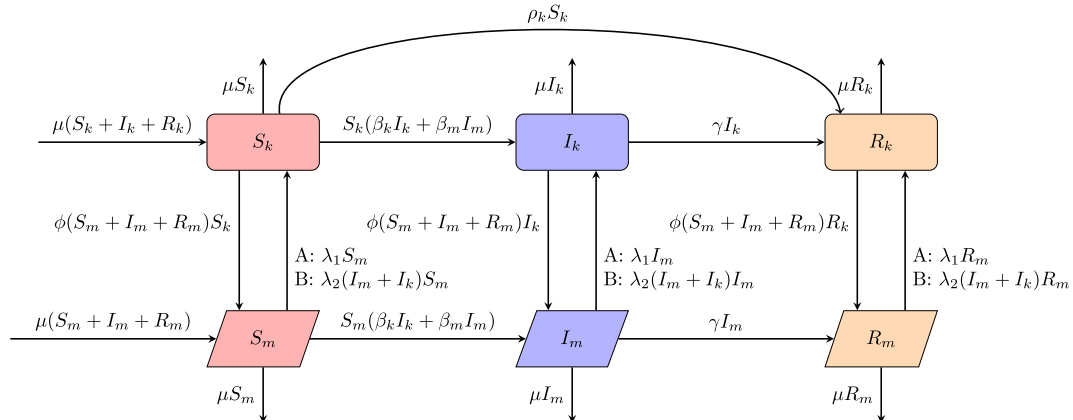


Fig. 3. Stock and flow diagram of the interaction between disease and misinformation. Misinformation “recovery” occurs either by simple constant recovery rate (A) or through contact with diseased individuals (B).

susceptible population. This is determined using the expression $R_i = \frac{\beta_i}{\gamma}$ for $i = k, m$ provided by (van den Driessche, 2017) for SIR compartmental models. Finally, the duration of misinformation infection under the simple misinformation recovery model is the inverse of the recovery rate, $\frac{1}{\lambda_1}$.

2.2. Calibration

The model was calibrated using weekly reported measles cases available from Health Canada for 2014 to 2024 (Public Health Agency of Canada, 2025) and the Public Health Ontario Enhanced Epidemiological Summary for measles (Public Health Ontario, 2025). These data were scraped and extracted using Python on 19 March 2025 and 5 May 2025, respectively. Only data from the most recent outbreak (1 December 2024 to 1 March 2025 for Health Canada and 30 December 2024 to 2 March 2025 for Public Health Ontario) were used for modelling.

Measles transmission coefficients for both information states β_m and β_k , misinformation transmission rate ϕ , and misinformation recovery rate λ were calibrated from calculated cumulative incidence using least-squares parameter estimation. This modelling was conducted in R using the deSolve (Soetaert et al., 2023) package and the LSODA algorithm to overcome ODE system stiffness (Petzold, 1983). Existing estimates of birth and death rates, vaccination rate, and disease recovery rate were fixed; disease transmission parameters and misinformation parameters were modelled (see Table 1 for details).

Model error was evaluated using root-mean-square error and visual inspection of the residual plot. Sensitivity analysis was conducted by determining predicted epidemic curves, final cumulative incidence, and peak incidence with each variable ($R_{0,k}$, $R_{0,m}$, ϕ , λ , $1/\gamma$, μ , and ρ_k) perturbed by $\pm 10\%$ while holding the other three at the best-fit or literature-estimated value.

3. Results

Parameters estimated from model fitting (β_m , β_k , ϕ , and λ), effective reproductive number for the knowledgeable and misinformed (R_m and R_k), misinformation recovery rate ($1/\lambda$), predicted outbreak statistics, and root-mean-square error are shown in Table 2 for the SMIRK model under both misinformation decay paradigms and both contexts (Canada and Ontario). For the three estimated parameters with ranges available from literature, nearly all deviate substantially from the expected.

Table 1
Table of parameters and literature estimates.

Symbol	Description and unit	Literature Estimate(s)
μ	Birth and death rate for the population of interest. Births/deaths per 1000 people.	10.4 (United Nations, 2025)
ρ_k	Disease vaccination rate among susceptible individuals. Unitless.	70–80 % (Ontario, 2023; Public Health Agency of Canada, 2024)
β_k	Disease transmission rate for knowledgeable infectious people. Weeks ^{−1} .	2.87×10^{-4} to 1.88×10^{-3} (Kuddus et al., 2021; Pokharel et al., 2024)
β_m	Disease transmission rate for misinformed infectious people. Weeks ^{−1} .	2.87×10^{-4} to 1.88×10^{-3} (Kuddus et al., 2021; Pokharel et al., 2024)
γ	Disease recovery rate. Weeks ^{−1} .	0.777 (McLean et al., 2013)
ϕ	Misinformation transmission rate. Weeks ^{−1} .	Insufficient literature to estimate.
λ	Misinformation recovery (simple, subscript 1) or knowledge transmission (infection-driven, subscript 2) rate. Weeks ^{−1} .	0.237 to 0.300 (Panchal et al., 2022)

Table 2
Table of parameters modelled and statistics derived using least-squares estimation on the SMIRK model for misinformation recovery which is constant ($\propto \lambda_1$) or proportional to the disease-infected population ($\propto \lambda_2 \sum I$) for Canada and Ontario.

Parameter	Description	Canada-wide		Ontario	
		$\propto \lambda_1$	$\propto \lambda_2 \sum I$	$\propto \lambda_1$	$\propto \lambda_2 \sum I$
β_k	Transmission rate for knowledgeable infectious	13.78	1.272×10^{-5}	1.196×10^6	2.712×10^{-3}
β_m	Transmission rate for misinformed infectious	1.167	1.159	1.350×10^{-5}	1.281
ϕ	Misinformation transmission rate	95.64	4.399×10^8	5.333×10^4	2.673×10^2
λ	Misinformation recovery rate	8.137×10^{-5}	1.421×10^6	0.0570	4.567×10^{-8}
R_k	Reproduction number for the knowledgeable	17.91	1.653×10^{-5}	1.555×10^6	3.526×10^3
R_m	Reproduction number for the misinformed	1.517	1.506	1.755×10^{-5}	1.666
$1/\lambda_1$	Duration of misinformation infection	1.229×10^4	–	17.54	–
max incidence		87	87	130	130
final cumulative incidence		973	978	1131	1128
RMSE		5.602	5.601	18.40	18.44

Predicted epidemic curves for measles are shown in Fig. 4A–D for Canada and Ontario, respectively for simple misinformation recovery (Fig. 3 scenario A and Equation (1)). Epidemic curves for contact-based misinformation “recovery” are omitted because incidence under both paradigms are substantially similar (see Fig. 5 in the Appendix).

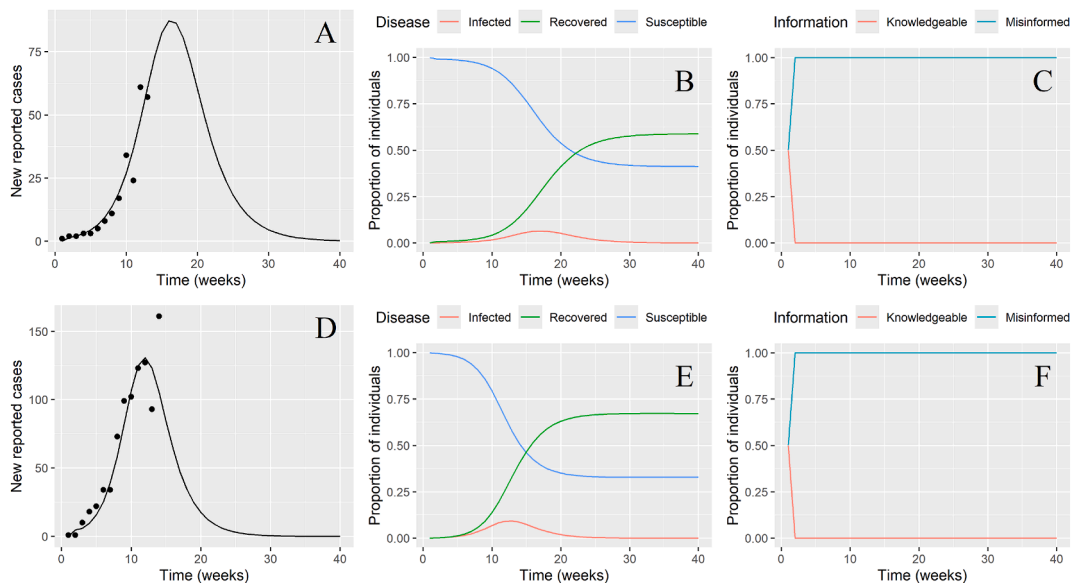


Fig. 4. Results of least-squares parameter estimation fitting the SMIRK model to cumulative incidence in Canada from 1 December 2024 to 1 March 2025 (top) and Ontario from 30 December 2024 to 2 March 2025 (bottom). (A, D) Actual (points) versus estimated and predicted cases (i.e., incidence). (B, E) Predicted disease state populations (susceptible, infected, and recovered) using model parameters. (C, F) Predicted information state populations (knowledgeable and misinformed) using model parameters.

The parameters given in Table 1 can be used to create SIR and KM curves (Fig. 4B and C for Canada and E and F for Ontario, respectively). For both Canadian and Ontario outbreaks, the SIR curves show a measles epidemic which infects only a portion of the population before settling to its equilibrium state. The KM curves, on the other hand, show misinformation immediately saturating the population. This is concordant with the stable equilibrium solution of the KM model with simple misinformation recovery using the estimated parameters given by $\frac{M}{K+M} = \left(1 - \frac{\lambda}{\phi}\right) \approx 1$.

Model RMSE values are presented in Table 2. Visual inspection of the residual plots (Appendix Fig. 6) shows that the SMIRK model is a decent fit for the Canadian measles outbreak, but predictive power is undermined at later time points for Ontario due to increased variance in new reported cases. Sensitivity analysis (Table 3 and Fig. 7 in the Appendix) revealed that the final epidemic curve for both misinformation recovery paradigms for Canada is highly sensitive to the reproduction number for the misinformed population $R_{0,m}$; sensitive to the disease recovery rate γ ; minimally sensitive to the misinformation rate ϕ and vaccination rate ρ_k ; and highly robust to changes in the misinformation recovery rate λ , the reproduction number for the knowledgeable population $R_{0,k}$, and the birth and death rate μ . For the two model paradigms applied to the Ontario outbreak, there is no such consistency in parameter sensitivity.

4. Discussion

Based on the results of this model, misinformation dynamics describable by the KMK model do not significantly impact infection dynamics of measles.

Fig. 4C and F indicate that misinformation is or will be pervasive as it quickly reaches its stable equilibrium with essentially the whole population misinformed. This is theoretically a dynamic equilibrium — though individuals still move back and forth between the knowledgeable and misinformed states, the net population transfer is zero. However, the extremely durable misinformation infection under the simple recovery paradigm (1.229×10^4 weeks (about 23.6 years) for Canada and 17.54 weeks for Ontario) and extreme infectiousness of misinformation (ϕ in Table 2) drive the high transmission relative to recovery rate of misinformation ($\frac{\phi}{\gamma}$) for both misinformation recovery paradigms and make the net movement functionally unidirectional. The durability of misinformation is likely impractical — misinformation is reversible over shorter time frames (Greyson & Bettinger, 2022; Tuckerman et al., 2022). Mathematically, the durability and infectiousness of misinformation are driven by the assumed high level of vaccine compliance among the knowledgeable population. This confirms the intuition that these outbreaks are not possible if a high proportion of individuals are vaccinated. This reinforces standard public health guidance that population education (and thereby vaccination) is key to preventing and mitigating these outbreaks.

Due to the rapid and durable spread of misinformation in the population, measles in this model behaves as a low- R_0 infection under the SIR model for both Canadian models and the infection-determined Ontario model ($R_m = 1.517, 1.506$, and 1.666 , respectively). This matches the sensitivity analysis for these cases, which reveals that predicted incidence is highly dependent on the reproduction number for the misinformed population and is relatively robust to other parameters (those concerning misinformation dynamics or governing the misinformed population). In contrast, the model with simple misinformation recovery for Ontario exhibits the inverse behavior, and is highly sensitive to multiple misinformation parameters and to R_k rather than R_m . This could indicate the existence of multiple error-minimizing parameter sets and overparameterization of the model.

There are a number of limitations of this model. Firstly, there are significant discrepancies in parameter estimates compared to existing transmission parameters for measles and misinformation duration. This is due in part to the differences between the SMIRK model presented here and those used in the literature. In particular, the measles transmission parameters were taken from papers employing compartmental models accounting for complex vaccination schedules and statuses (Kuddus et al., 2021; Pokharel et al., 2024). The misinformation transmission parameter is estimated from an infection model which also accounts for those exposed to misinformation and misinformed “hibernators” (an SEHIR model) (Panchal et al., 2022).

Secondly, this model of misinformation is limited and makes potentially invalid assumptions. The KMK model with simple misinformation recovery has two equilibria — one stable and one unstable — and thus will never oscillate. This is inconsistent with general understanding of misinformation dynamics (Panchal et al., 2022) and precludes substantial forcing of disease infection in the composite model. Though contact-based misinformation recovery theoretically permits oscillation, no oscillations are evident here.

Thirdly, the SMIRK model composite was constructed in the simplest, most naive way. For instance, the assumption of identical misinformation transmission dynamics between disease states is likely to be invalid. As an example, measles infection may be a catalyst for individuals to be treated by a doctor, who may simultaneously disabuse them of their misinformation. The model also assumes other dimensions of homogeneity within each compartment, in particular age and explicit vaccination status as has been found to be useful elsewhere (Kuddus et al., 2021; Pokharel et al., 2024). Finally, the model assumes that misinformation and measles operate on similar timescales. However, as suggested by this model, it is

possible that misinformation is much more durable than measles or that the effects of misinformation on disease are associated with dynamics other than misinformation spread.

Finally, there are limited existing comparisons, parameters, and data for misinformation models. Without knowledge of how misinformation differs across contexts, it is difficult to validate the misinformation model and parameters presented here.

Beyond model limitations, the parameterization presented here is only as reliable as the data used to calibrate it. With rapidly developing outbreaks and necessarily imperfect reporting, prediction of epidemic peaks and size remains challenging, though this problem is not limited to the SMIRK model presented here. Future applications could utilize corrected historical data or estimated inflated real-time data to improve prediction accuracy.

The numerous limitations open up opportunities for further research, namely, to refine and validate the misinformation model. Future research could use more complex compartments or composition, for example, a naive - knowledgeable - misinformed - disinformed model which would allow for more complex oscillations. Alternately, a network-based model could be composed of social media and in-person social network data to model the spatial dynamics of the disease in space and cyberspace. Because this exploits the inherent network structure of social media — where most people are infected with misinformation — the dynamics may more closely match reality.

Future validation of misinformation models should incorporate public survey data on vaccine perception (e.g., MMR vaccine hesitancy data ([Public Health Agency of Canada, 2024](#))) and other indicators of public sentiment and understanding.

Though the model presented here does not significantly add to understanding of how misinformation affects disease transmission dynamics, its design is a useful proof of concept for future combined models. These models can be used to further develop knowledge of how micro-level misinformation transmission affect infection, adding to well-established macro-level understanding. By interrogating the interplay of information and disease transmission, we can develop more effective interventions to combat both biological and informational epidemics.

CRediT authorship contribution statement

Callandra Moore: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **David Fisman:** Writing – review & editing, Supervision.

Declaration of conflict interest

CM declares no conflicts of interest related to this manuscript. DF has served on advisory boards for Seqirus and Pfizer vaccines. The research was supported by grants to DF (CIHR COVID-19 rapid research funding OV4-170360 and CIHR operating grant #518192).

Appendix

A Supplemental Exhibits

$$\begin{aligned}
 \frac{dS_k}{dt} &= -\rho_k S_k - S_k(\beta_k I_k + \beta_m I_m) - \phi(S_m + I_m + R_m)S_k + \lambda_1 S_m + \mu(S_k + I_k + R_k) - \mu S_k \\
 \frac{dI_k}{dt} &= S_k(\beta_k I_k + \beta_m I_m) - \gamma I_k - \phi(S_m + I_m + R_m)I_k + \lambda_1 I_m - \mu I_k \\
 \frac{dR_k}{dt} &= \rho_k S_k + \gamma I_k - \phi(S_m + I_m + R_m)R_k + \lambda_1 R_m - \mu R_k \\
 \frac{dS_m}{dt} &= -S_m(\beta_k I_k + \beta_m I_m) + \phi(S_m + I_m + R_m)S_k - \lambda_1 S_m + \mu(S_m + I_m + R_m) - \mu S_m \\
 \frac{dI_m}{dt} &= S_m(\beta_k I_k + \beta_m I_m) - \gamma I_m + \phi(S_m + I_m + R_m)I_k - \lambda_1 I_m - \mu I_m \\
 \frac{dR_m}{dt} &= \gamma I_m + \phi(S_m + I_m + R_m)R_k - \lambda_1 R_m - \mu R_m
 \end{aligned} \tag{1}$$

$$\begin{aligned}
\frac{dS_k}{dt} &= -\rho_k S_k - S_k(\beta_k I_k + \beta_m I_m) - \phi(S_m + I_m + R_m) S_k + \lambda_2(I_m + I_k) S_m + \mu(S_k + I_k + R_k) - \mu S_k \\
\frac{dI_k}{dt} &= S_k(\beta_k I_k + \beta_m I_m) - \gamma I_k - \phi(S_m + I_m + R_m) I_k + \lambda_2(I_m + I_k) I_m - \mu I_k \\
\frac{dR_k}{dt} &= \rho_k S_k + \gamma I_k - \phi(S_m + I_m + R_m) R_k + \lambda_2(I_m + I_k) R_m - \mu R_k \\
\frac{dS_m}{dt} &= -S_m(\beta_k I_k + \beta_m I_m) + \phi(S_m + I_m + R_m) S_k - \lambda_2(I_m + I_k) S_m + \mu(S_m + I_m + R_m) - \mu S_m \\
\frac{dI_m}{dt} &= S_m(\beta_k I_k + \beta_m I_m) - \gamma I_m + \phi(S_m + I_m + R_m) I_k - \lambda_2(I_m + I_k) I_m - \mu I_m \\
\frac{dR_m}{dt} &= \gamma I_m + \phi(S_m + I_m + R_m) R_k - \lambda_2(I_m + I_k) R_m - \mu R_m
\end{aligned} \tag{2}$$

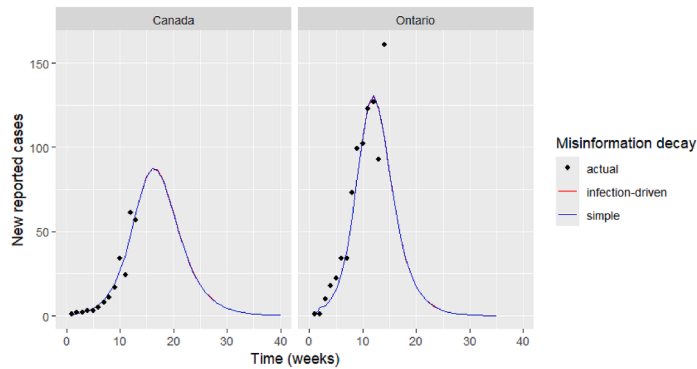


Fig. 5. Comparison of extrapolated epidemic curves applying the SMIRK model with simple constant recovery rate or through contact with disease-infected individuals for the Canadian and Ontario measles outbreaks.

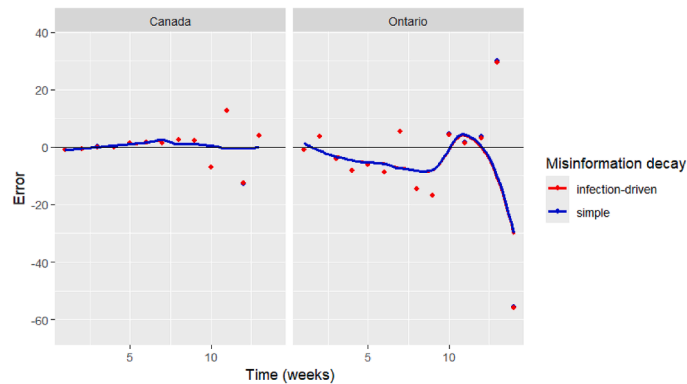


Fig. 6. Residual plot of estimated versus predicted incident measles cases for Ontario and Canadian measles outbreaks. Smoothed line is local regression (LOESS) interpolation. Incidence estimated using the SMIRK model with simple constant recovery rate or through contact with disease-infected individuals.

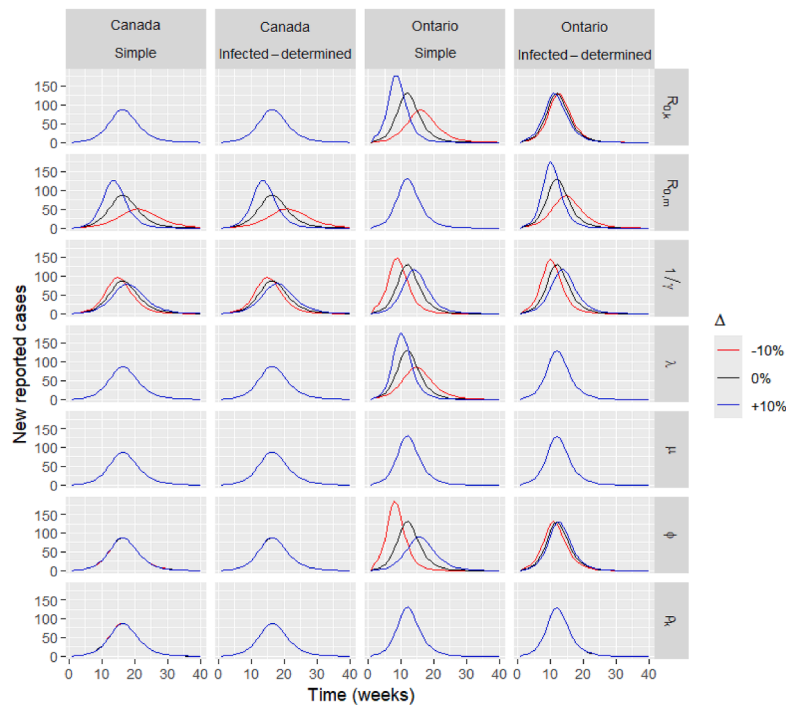


Fig. 7. Predicted incidence, perturbing each variable by $\pm 10\%$ from the SMIRK model with simple and infection-determined misinformation recovery for both Canada and Ontario. Baseline ($\Delta 0\%$) parameter values shown in Table 1.

Table 3
Maximum incidence and final cumulative incidence (in parentheses) varying each modelled parameter by $\pm 10\%$. Baseline ($\Delta 0\%$) parameter values shown in Table 1. At baseline, from left to right and top to bottom, incidence and cumulative incidence are 87 and 973, 87 and 978, 130 and 1131, and 130 and 1128 cases, respectively.

		Simple		Infected-determined	
		−10 %	+10 %	−10 %	+10 %
Canada	$R_{0,k}$	87 (973)	87 (973)	87 (978)	87 (978)
	$R_{0,m}$	50 (774)	126 (1116)	51 (778)	126 (1119)
	ϕ	86 (972)	87 (974) 87	(978)	87 (978)
	λ	87 (973)	87 (973)	87 (978)	87 (978)
	μ	87 (973)	87 (973)	87 (978)	87 (978)
	ρ_k	87 (974)	86 (971)	87 (978)	87 (978)
	$1/\gamma$	97 (975)	79 (971)	97 (977)	79 (975)
Ontario	$R_{0,k}$	85 (968)	176 (1254)	128 (1126)	130 (1128)
	$R_{0,m}$	130 (1131)	130 (1131)	85 (966)	177 (1247)
	ϕ	185 (1265)	89 (986)	130 (1127)	128 (1129)
	λ	85 (967)	177 (1252)	130 (1128)	130 (1128)
	μ	130 (1131)	130 (1131)	130 (1128)	130 (1129)
	ρ_k	130 (1131)	130 (1131)	130 (1128)	129 (1129)
	$1/\gamma$	147 (1137)	118 (1127)	145 (1131)	117 (1128)

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